

Chuan Qin

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Objective: Embodied AI Algorithm Intern (VLA / Robot Learning) · Available immediately · 6+ months internship



Links

- Homepage: <https://chuanqin8.github.io/>
- Google Scholar: <https://scholar.google.com/citations?user=17NMHvUAAAAJ> (Citations: 93, h-index: 2, i10-index: 2)
- GitHub: <https://github.com/chuanqin8>

Education

Lanzhou University (985 Project) Ph.D. in Computer Application Technology 2021 – 2027 (expected)
Harbin Institute of Technology (985 Project, C9) B.Eng., Naval Architecture & Ocean Engineering 2017 – 2021

Projects

Tabletop Object-Organizing Embodied AI System (StarVLA · Franka) Real-robot · Spatial generalization · Data loop

- **Scenario:** autonomous recognition, grasping, and organizing of target objects on a tabletop (pick-and-place into a box, sequential stacking of colored blocks, etc.); independently delivered the full pipeline from data collection and model training to real-robot deployment and data-loop iteration.
- **Data collection:** pylibfranka requires a real-time kernel unavailable on ordinary IPCs; with AI-assisted programming, built a pylibfranka-based teleoperation SDK that works without a real-time kernel, synchronously recording multi-view RGB images, proprioceptive states, and end-effector poses at 20 Hz (LeRobot v3 format).
- **Training:** post-trained a VLA model on the StarVLA framework (Qwen3-VL-4B + flow-matching action head); introduced a dedicated gripper classification head and state-dropout regularization for delicate manipulation, significantly improving target-following and success rates in real-robot tests.
- **Deployment:** the policy runs real-time closed-loop inference at the edge; with asynchronous inference and action-chunk smoothing, it sustains smooth continuous 20 Hz motion at a measured 115 ms per inference; stably completes pick-and-place-into-box under arbitrary poses of the object and the container; multiple training-/deployment-side root causes were reverse-located and fixed from real-world failure cases.
- **Data loop:** reproduced the RECAP mechanism from $\pi^*0.6$ (per-frame advantage from a value net + advantage-conditioned training), combined with DAgger correction, closing the "deploy → corrective collection → automatic retraining" data flywheel; after corrective iterations, the success rate improved notably under arbitrary poses.

Robust Post-Training of Embodied AI Models (StarVLA · LIBERO) Ongoing · Paper in preparation

- VLA models degrade sharply under out-of-distribution disturbances after fine-tuning; this project investigates robust post-training strategies from the optimization side, reproducing and slightly surpassing the official strong baseline (97.2% vs 96.6%), with consistent gains across multiple physical perturbation axes of LIBERO-Plus—camera viewpoint, initial arm pose, lighting, and more—confirmed by multi-seed reproduction.

Out-of-Distribution Robustness of World Action Models (LIBERO / LIBERO-Plus) Ongoing · Paper in preparation

- World action models degrade substantially under visual disturbances; this project systematically analyzes their internal mechanisms and failure modes, with large-scale evaluations across the multi-axis perturbations of LIBERO-Plus, informing the design of low-cost robustness methods.

Survey of LLM-Based AI Agents Tsinghua Science and Technology · In Press

- A holistic review deconstructing six core modules of LLM-based agents—reasoning, perception, memory, planning, action, and learning—and their interplay; surveyed the application spectrum from digital domains to embodied AI systems, identifying key challenges and **providing a roadmap for next-generation embodied agents.**

Efficient Sharpness-Aware Optimization: SAM-LA / SAM-L IEEE/CAA JAS Accepted · ICLR 2027 Submitted

- To address SAM's roughly 2x training overhead, proposed loss acceleration (LA) and loss gap— $O(1)$ flatness measures that directly characterize loss-landscape sharpness along the optimization trajectory under gradient flow, replacing computationally prohibitive Hessian-based metrics.
- Built optimizers that adaptively switch between SAM and SGD (SAM only in sharp regions, efficient SGD elsewhere), achieving comparable or better generalization with substantially lower training cost on CIFAR/ImageNet, with convergence analysis.
- **Targeting embodied AI's core need for stronger generalization at lower cost, these methods transfer directly to VLA post-training and real-robot policy fine-tuning, alleviating the out-of-distribution generalization bottleneck.**

Adaptive Gradient Modulation: GAF / AGR / FedAvg-AGA Neural Networks 2023 · IEEE TCE 2025 · TNNLS Minor Revision

- Proposed gradient activation functions that adaptively reshape gradients—amplifying small components and constraining large ones—alleviating vanishing/exploding gradient and ill-conditioned problems and accelerating deep LSTM convergence; 81 Google Scholar citations with open-sourced code.
- Revealed SAM's escape mechanism through gradient statistics (fewer small elements, larger variance and norm) and proposed Adaptive Gradient Reshaping (AGR), which progressively enlarges minor gradient elements and benefits both SGD and SAM; Hessian max-eigenvalue and trace analyses confirm convergence to flatter minima.
- Validated on large-model training: on GPT-2, under identical hyperparameters versus the AdamW baseline, training loss reduced by 10.8%, test loss by 9.1%, and perplexity by 14.1%, meeting the acceptance criteria of the CCF-Huawei Populus Euphratica Fund.
- Introduced gradient activation into client-side federated learning (FedAvg-AGA / FedSAM-AGA), mitigating training instability and overfitting on non-IID consumer-electronics data and improving federated learning performance, with convergence analysis.
- **For embodied AI—where real-robot data is scarce and post-training budgets are limited—these gradient modulation methods provide optimizer-level support for stable and efficient post-training of LLM and VLA models, and transfer directly to real-robot policy fine-tuning, ensuring stable convergence and generalization in few-shot scenarios.**

- FSCNO: sign-based momentum dynamics for precise local search plus a fuzzy controller that adapts the learning rate in real time to population diversity; **validated in motion-control experiments on a soft manipulator and a redundant rigid manipulator**.
- STPAN: self-tuning perturbation-aware neurodynamics for dynamic multilinear quaternion tensor equations, identifying and adaptively compensating disturbances online; applied to redundant-manipulator trajectory tracking.
- CNOGNP: collaborative neurodynamic optimization with a gradient-norm-penalized fitness function, using the low-cost gradient norm as a sharpness proxy to efficiently locate flat minima; classification experiments confirm improved generalization.
- QICNO: quantum-tunneling-like local search via gradient transformation plus global search with dynamic attractors and stochastic jumps; outperforms mainstream optimizers in accuracy and convergence speed on four traffic-flow benchmarks for in-vehicle navigation and other consumer-electronics scenarios.

ODE-Inspired Neural Network Architecture Design

IEEE TETCI / TNNLS Under Review · NMI Submitted

- Interpreting residual forward propagation as Euler discretization of ODEs, established the correspondence between the truncation error of Adams–Bashforth multi-step discretizations and training loss, proposing that numerical consistency is key to fitting ability and degradation alleviation; validated on CIFAR10/100 and ImageNet.
- From the sampling-theorem perspective, proposed approximate sampling and network forward sampling frequency, and designed Taylor k-step (TLk) connections to replace residual connections; experiments show that satisfying approximate sampling and increasing the sampling frequency consistently improve ResNet and ViT performance and flatten the loss landscape.
- Further proposed a ternary duality hypothesis linking dynamical systems, information theory, and deep learning, unifying global/local error laws across the three systems and offering a new framework for understanding neural networks.
- Unifying the understanding of neural networks from a dynamical-systems perspective, which naturally aligns with robot dynamics modeling and provides a theoretical basis for co-designing network architectures and control dynamics in embodied agents.

Grants

- Generalization of Deep Neural Networks via Neurodynamics — **"Star of Innovation" Graduate Program of Gansu Province**, CNY 10,000, **Principal Investigator** (sole proposal author; independently responsible for all research work) Oct. 2025 – Oct. 2026
- Differential-Equation Interpretation of LLM Generalization and Robustness — **CCF-Huawei Populus Euphratica Fund (Theoretical Computer Science)**, CNY 110,000, Key Participant (sole author of proposal, mid-term, and closing materials; led research design and experimental validation) Oct. 2024 – Oct. 2025
- Stability, Convergence, and Optimization of Differential-Equation-Interpreted LLMs on PaddlePaddle — **CCF-Baidu Pinecone Fund**, CNY 150,000, Key Participant (sole proposal author; managed the full cycle from application to final acceptance) Aug. 2023 – Jul. 2024
- Differential-Equation-Inspired Network Architecture and Optimizer Design on MindSpore — **CAAI-Huawei MindSpore Academic Reward Fund**, CNY 90,000, Key Participant (sole proposal author; contributed to core algorithm design and experimental validation) Nov. 2022 – Nov. 2023

Skills

- Programming & tools: Python, PyTorch, Linux.
- AI-assisted R&D: familiar with vibe coding / vibe research workflows; independently delivered engineering tasks such as a data-collection SDK via AI coding tools.
- LLM & embodied AI: VLA / WAM architectures; SFT post-training.
- Robotics: real-robot deployment and teleoperation data collection on Franka manipulators (self-developed data-collection SDK).

Honors & Awards

- First Prize, Annual Conference Paper Award
- First-Class Academic Scholarship
- Outstanding Graduate Student, Lanzhou University

Publications

- Chuan Qin and Long Jin, "A Survey of LLM-Based AI Agents," *Tsinghua Science and Technology*, **in press**, DOI: 10.26599/TST.2026.9010072.
- Chuan Qin, Liangming Chen, Zangtai Cai, Mei Liu, and Long Jin, "Long short-term memory with activation on gradient," *Neural Networks*, vol. 164, pp. 135–145, 2023. **(SCI Q1 Top at publication; 81 Google Scholar citations)**
- Chuan Qin, Mei Liu, and Long Jin, "Adaptive gradient activation for federated learning on consumer electronic devices," *IEEE Transactions on Consumer Electronics*, vol. 71, no. 2, pp. 5489–5500, 2025. **(IEEE Transactions journal)**
- Chuan Qin, Liangming Chen, and Long Jin, "SAM-LA: Efficient sharpness-aware minimization training via loss acceleration," *IEEE/CAA Journal of Automatica Sinica*, 2026, DOI: 10.1109/JAS.2026.126491. **(Accepted, SCI Q1 Top, IF 19.2)**
- Chuan Qin, Liangming Chen, and Long Jin, "Adaptive gradient reshaping boosting generalization," *IEEE Transactions on Neural Networks and Learning Systems*, **minor revision**.
- Chuan Qin and Long Jin, "Anti-perturbation solution of dynamic multilinear quaternion tensor equations with application to manipulators," *CAAI Transactions on Intelligence Technology*, **major revision**.
- Chuan Qin and Long Jin, "Collaborative neurodynamic optimization with gradient norm penalty for generalization improvement," *IEEE Transactions on Cybernetics*, **reject & resubmit**.
- Chuan Qin and Long Jin, "Fuzzy sign-based collaborative neurodynamic optimization with application to robots," *IEEE Transactions on Fuzzy Systems*, **under review**.
- Chuan Qin, Liangming Chen, Yongji Guan, and Long Jin, "SAM-L: Dynamically balancing sharpness-awareness and efficiency with loss gap metric," *ICLR 2027*, **under review**.
- Chuan Qin, Jianfeng Lv, and Long Jin, "Quantum-inspired collaborative neurodynamic optimization for traffic flow prediction," *IEEE Transactions on Consumer Electronics*, **under review**.
- Chuan Qin, Liangming Chen, and Long Jin, "Consistency is key of fitting and degradation alleviating for deep neural networks," *IEEE Transactions on Neural Networks and Learning Systems*, **under review**.
- Chuan Qin, Mei Liu, and Long Jin, "Understanding residual neural networks from sampling theorem perspective," *IEEE Transactions on Emerging Topics in Computational Intelligence*, **under review**.

13. Chuan Qin, Long Jin, and Zhigang Zeng, "On ternary duality of dynamical systems, information theory, and deep learning," *NMI*, submitted.

Others

- Half-marathon finisher: 2024 Lanzhou Marathon (01:42:38); 2024 Yingshan Marathon (01:53:39)

Publication Details

A Survey of LLM-Based AI Agents

Tsinghua Science and Technology / In Press

Chuan Qin, Long Jin

- **Abstract:** We provide a holistic review of LLM-based AI agents, deconstructing the core architecture—reasoning, perception, memory, planning, action, and learning—and examining the interplay among these modules. We further survey their transformative impact across applications from digital domains to embodied systems, and conclude with critical challenges and future directions as a roadmap for next-generation LLM-based agents.

Long Short-Term Memory with Activation on Gradient

Neural Networks, 2023 / Citation: 81 (Google Scholar)

Chuan Qin, Liangming Chen, Zangtai Cai, Mei Liu, Long Jin

- **Abstract:** We apply gradient activation—modifying gradients with a designed gradient activation function—to LSTM training, and provide empirical criteria for choosing its hyperparameters. The method alleviates the vanishing/exploding gradient and ill-conditioned problems and accelerates the convergence of LSTM.
- **Code:** <https://github.com/LongJin-lab/ACT-In-NLP>

Adaptive Gradient Activation for Federated Learning on Consumer Electronic Devices

IEEE Transactions on Consumer Electronics, 2025 / Citation: 10 (Google Scholar)

Chuan Qin, Mei Liu, Long Jin

- **Abstract:** We propose FedAvg-AGA and FedSAM-AGA, which introduce adaptive gradient activation into client-side federated training on consumer electronic devices. The methods alleviate vanishing/exploding gradients, help client models escape sharp minima under non-IID data, and improve federated learning performance, with convergence analysis provided.

SAM-LA: Efficient Sharpness-Aware Minimization Training via Loss Acceleration

IEEE/CAA Journal of Automatica Sinica, 2026 / Accepted · SCI Q1 Top · Impact Factor 19.2

Chuan Qin, Liangming Chen, Long Jin

- **Abstract:** We propose loss acceleration (LA), an $O(1)$ sharpness measure that directly captures the loss landscape's sharpness along the optimization trajectory under the gradient flow setting. Building on LA, SAM-LA adaptively alternates between SAM and SGD, achieving generalization comparable to or better than SAM while substantially reducing training time and computational overhead, with convergence analysis provided.

Adaptive Gradient Reshaping Boosting Generalization

IEEE Transactions on Neural Networks and Learning Systems / Minor Revision

Chuan Qin, Liangming Chen, Long Jin

- **Abstract:** We demonstrate that SAM gradients contain fewer small elements and exhibit considerably greater variance and norm than SGD gradients, suggesting an enhanced ability to circumvent sharp minima. Inspired by this, adaptive gradient reshaping (AGR) progressively enlarges minor gradient elements, boosting the sharp-minima-escaping ability of both SGD and SAM, corroborated by Hessian trace/max-eigenvalue and loss-landscape analyses on CIFAR and ImageNet.

Anti-Perturbation Solution of Dynamic Multilinear Quaternion Tensor Equations With Application to Manipulators

CAAI Transactions on Intelligence Technology / Major Revision

Chuan Qin, Long Jin

- **Abstract:** We devise a self-tuning perturbation-aware neurodynamics (STPAN) method for solving dynamic multilinear quaternion tensor equations under perturbations. The solver identifies the temporal profile of disturbances on the fly and adaptively offsets them; convergence and disturbance-estimating capacity are formally established, with numerical case studies and a redundant-manipulator trajectory-tracking task confirming its practical merit.

Collaborative Neurodynamic Optimization with Gradient Norm Penalty for Generalization Improvement

IEEE Transactions on Cybernetics / Reject & Resubmit

Chuan Qin, Long Jin

- **Abstract:** We introduce collaborative neurodynamic optimization with a gradient-norm-penalized fitness function (CNOGNP), which efficiently finds flat minima using the low-cost gradient norm as a sharpness proxy. Theoretical analysis and simulations demonstrate effective flat-minima seeking, and classification experiments validate improved generalization over competing algorithms.

Fuzzy Sign-Based Collaborative Neurodynamic Optimization With Application to Robots

IEEE Transactions on Fuzzy Systems / Under Review

Chuan Qin, Long Jin

- **Abstract:** We propose fuzzy sign-based collaborative neurodynamic optimization (FSCNO), integrating sign-based momentum dynamics for precise local search with a collaborative global-exploration mechanism mediated by a fuzzy logic controller that adapts the learning rate to real-time population diversity and optimization progress. Evaluations on high-dimensional benchmarks and kinematic control of a soft robot and a redundant rigid robot—in both simulations and physical experiments—validate its efficacy.

SAM-L: Dynamically Balancing Sharpness-Awareness and Efficiency With Loss Gap Metric

ICLR 2027 / Submitted

Chuan Qin, Liangming Chen, Yongji Guan, Long Jin

- **Abstract:** We propose SAM-L, which dynamically switches between SAM and SGD based on the loss gap—a computationally inexpensive $O(1)$ flatness measure approximating the gradient norm. SAM-L significantly reduces computational overhead by cutting the number of SAM iterations while achieving comparable or even superior generalization, with convergence analysis and evaluations on CIFAR and ImageNet.

Quantum-Inspired Collaborative Neurodynamic Optimization for Traffic Flow Prediction

IEEE Transactions on Consumer Electronics / Under Review

Chuan Qin, Jianfeng Lv, Long Jin

- **Abstract:** We introduce quantum-inspired collaborative neurodynamic optimization (QICNO), combining a local search whose nonlinear gradient transformation acts analogously to quantum tunneling with a global search guided by dynamically computed attractors and stochastic jumps. On four benchmark traffic datasets with several forecasting models, QICNO consistently outperforms standard optimization algorithms in both accuracy and convergence speed.

Consistency Is Key of Fitting and Degradation Alleviating for Deep Neural Networks

IEEE Transactions on Neural Networks and Learning Systems / Under Review

Chuan Qin, Liangming Chen, Long Jin

- **Abstract:** We connect the consistency in numerical analysis with neural networks' performance: viewing a residual network as the 1-step Adams-Bashforth discretization of an ODE, we find the discretization's truncation error corresponds to the network loss, making consistency crucial to fitting ability and degradation alleviation. Experiments on CIFAR10, CIFAR100, and ImageNet confirm the finding, and show that without consistency networks are more prone to vanishing/exploding gradients and feature oscillation.
- **Code:** <https://github.com/LongJin-lab/consistency>

Understanding Residual Neural Networks From Sampling Theorem Perspective

IEEE Transactions on Emerging Topics in Computational Intelligence / Under Review

Chuan Qin, Mei Liu, Long Jin

- **Abstract:** We bridge residual connections and network performance from the perspective of the sampling theorem: when a network's discretization satisfies approximate sampling, increasing the network forward sampling frequency improves performance. We design Taylor k-step (TLk) connections extending residual connections, and experiments on TLk networks and vision Transformers show that ensuring approximate sampling guarantees trainability while increasing the frequency enhances the loss landscape's flatness, which is related to generalization.

On Ternary Duality of Dynamical Systems, Information Theory, and Deep Learning

NMI / Submitted

Chuan Qin, Long Jin, Zhigang Zeng

- **Abstract:** We propose a ternary duality hypothesis linking dynamical systems, information theory, and deep learning under a consistent framework of using discrete states to approximate exact continuous ones. Three predictions—on global errors, optimal intervals, and local-error effects—are supported by empirical phenomena across all three systems, offering a unified lens for understanding deep neural networks.